

HW 1

Problem 1

1.

$$\{(1, 1), (1, -1)\}$$

2.

$$\{(1, -1, 1), (1, -1, -1), (-1, 1, 1), (-1, 1, -1)\}$$

3.

$$A_n = \{(a_1, a_2, \dots, a_n, \dots) \in \Omega \mid \sum_{i=1}^n a_i = 0\}$$

4.

$$\bigcup_{n=1}^{\infty} A_n$$

5.

No for example the infinite sequence of +1's

Problem 2

1.

Since $\Omega = \bigcup_{n \in N} \{n\}$ by definition

$$\begin{aligned}\mathbb{P}(\Omega) &= \sum_{n \in N} \mathbb{P}(\{n\}) = 1 \\ \sum_{n \in N} C_s n^{-s} &= 1 \\ C_s \sum_{n \in N} n^{-s} &= 1 \\ C_s &= \frac{1}{\sum_{n \in N} n^{-s}}\end{aligned}$$

2.

Since $A = \{2, 4, 6, \dots\} = \bigcup_{n \in N} \{2n\}$ by definition

$$\begin{aligned}\mathbb{P}(A) &= \sum_{n \in N} \mathbb{P}(\{2n\}) \\ &= \sum_{n \in N} C_s (2n)^{-s} \\ &= C_s 2^{-s} \sum_{n \in N} n^{-s}\end{aligned}$$

From 1. we know $C_s = \frac{1}{\sum_{n \in N} n^{-s}}$

$$= \frac{1}{2^s}$$

3.

Since $A = \{q, 2q, 3q, 4q, \dots\} = \bigcup_{n \in N} \{qn\}$ by definition

$$\begin{aligned}\mathbb{P}(A) &= \sum_{n \in N} \mathbb{P}(\{pn\}) \\ &= \sum_{n \in N} C_s (pn)^{-s} \\ &= C_s p^{-s} \sum_{n \in N} n^{-s}\end{aligned}$$

From 1. we know $C_s = \frac{1}{\sum_{n \in N} n^{-s}}$

$$= \frac{1}{p^s}$$

Do the same thing for B and we get $\mathbb{P}(B) = \frac{1}{q^s}$

Do the same thing for $A \cap B = \{pq, 2pq, 3pq, \dots\}$ and we get $\mathbb{P}(A \cap B) = \frac{1}{(pq)^s}$

$$\mathbb{P}(A)\mathbb{P}(B) = \mathbb{P}(A \cap B) = \frac{1}{(pq)^s}$$

Thus by definition event A and event B are independent with respect to this measure $\mathbb{P}$

Problem 3

1.

First consider $A \cap B = (A^c \cup B^c)^c$

$$\begin{aligned}\mathbb{P}(A \cap B) &= \mathbb{P}((A^c \cup B^c)^c) \\ &= 1 - \mathbb{P}(A^c \cup B^c)\end{aligned}$$

Proven in class $\mathbb{P}(A^c \cup B^c) \leq \mathbb{P}(A^c) + \mathbb{P}(B^c)$

$$\begin{aligned}&\geq 1 - \mathbb{P}(A^c) - \mathbb{P}(B^c) \\ &= 1 - 1 + \mathbb{P}(A) - 1 + \mathbb{P}(B) \\ &= \mathbb{P}(A) + \mathbb{P}(B) - 1\end{aligned}$$

Thus $\mathbb{P}(A \cap B) \geq \mathbb{P}(A) + \mathbb{P}(B) - 1$

Now consider $(A \cap B) \cap C$ apply the proof above to these sets.

$$\mathbb{P}((A \cap B) \cap C) \geq \mathbb{P}(A \cap B) + \mathbb{P}(C) - 1$$

Now apply proof again

$$\geq \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - 2$$

Therefore $\mathbb{P}((A \cap B) \cap C) \geq \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - 2$

2.

Collaborators: Keertan

Let $S = \{1, \dots, 100\}$ and $\mathbb{P}(D) = |D|/100$ for $D \subset S$

$A = \{1, 2, 3, \dots, 70\}$

$B = \{1, 2, 3, \dots, 45, 71, 72, \dots, 100\}$

$C = \{1, 2, 3, \dots, 25, 46, 47, \dots, 100\}$

$|A| = 70, |B| = 75, |C| = 80$

$A \cap B \cap C = \{1, 2, \dots, 25\}$

Thus $\mathbb{P}(A \cap B \cap C) = .25$

Thus the minimal value of $\mathbb{P}(A \cap B \cap C) = .25$

Problem 4

$$B_1 = \{1, 2, 3, 4, 5, 6\}$$

$$B_2 = \{2, 4, 6\}$$

$$B_3 = \{3, 6\}$$

1.

$$\mathbb{P}(6) = \mathbb{P}(6 \mid B_1)P(B_1) + \mathbb{P}(6 \mid B_2)P(B_2) + \mathbb{P}(6 \mid B_3)P(B_3) = \frac{1}{3} \left(\frac{1}{6} + \frac{1}{3} + \frac{1}{2} \right) = \frac{1}{3}$$

2.

$$\mathbb{P}(B_2 \mid 6) = \frac{\mathbb{P}(B_2 \cap 6)}{\mathbb{P}(6)} = \frac{\mathbb{P}(6 \mid B_2)\mathbb{P}(B_2)}{\frac{1}{3}} = \frac{\frac{1}{3} \times \frac{1}{3}}{\frac{1}{3}} = \frac{1}{3}$$

3.

$$\begin{aligned} \mathbb{P}(6 \text{ on second} \mid 6 \text{ on first}) &= \frac{\mathbb{P}(6 \text{ on second} \cap 6 \text{ on first})}{\mathbb{P}(6)} = \frac{\frac{1}{3}(\mathbb{P}(6 \mid B_1)^2 + \mathbb{P}(6 \mid B_2)^2 + \mathbb{P}(6 \mid B_3)^2)}{\frac{1}{3}} \\ &= \left(\frac{1}{36} + \frac{1}{9} + \frac{1}{4} \right) = \frac{7}{18} \end{aligned}$$

4.

$$\mathbb{P}(B_2 \mid \text{diff}) = \frac{\mathbb{P}(B_2 \cap \text{diff})}{\mathbb{P}(\text{diff})} = \frac{\mathbb{P}(\text{diff} \mid B_2)\mathbb{P}(B_2)}{\mathbb{P}(\text{diff})} = \frac{\frac{2}{3} \times \frac{1}{3}}{\frac{1}{3}(\mathbb{P}(\text{diff} \mid B_1) + \dots)} = \frac{\frac{2}{3}}{(\frac{5}{6} + \frac{2}{3} + \frac{1}{2})} = \frac{1}{3}$$

5.

$$\begin{aligned}
 \mathbb{P}(\text{Same from two diff bags}) &= \\
 &= \mathbb{P}(\text{Same from two diff} \mid B_1, B_2) \mathbb{P}(B_1, B_2) \\
 &+ \mathbb{P}(\text{Same from two diff} \mid B_1, B_3) \mathbb{P}(B_1, B_3) \\
 &+ \mathbb{P}(\text{Same from two diff} \mid B_2, B_3) \mathbb{P}(B_2, B_3)
 \end{aligned}$$

$$\mathbb{P}(\text{Same from two diff} \mid B_1, B_2) = \frac{\text{number of good outcomes}}{\text{all outcomes}} = \frac{3}{18} = \frac{1}{6}$$

$$\mathbb{P}(\text{Same from two diff} \mid B_1, B_3) = \frac{2}{12} = \frac{1}{6}$$

$$\mathbb{P}(\text{Same from two diff} \mid B_2, B_3) = \frac{1}{6}$$

$$= \frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$$

6.

$$\mathbb{P}(B_1 B_2 \mid \text{Same from two diff}) = \frac{\mathbb{P}(\text{Same from two diff} \mid B_1, B_2) \mathbb{P}(B_1 B_2)}{\mathbb{P}(\text{Same from two diff bags})} = \frac{1}{3}$$